

MAT31 - 1

Linear Algebra III — Euclidean Spaces

[Module page](#)

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About this textbook

This year, for the first time, every Bachelor course at Albert School has its own textbook, thanks to your teachers and AI.

- The textbook is a help for working on the course at home, after the class.
- The course as taught in class remains the reference, and prevails over the textbook wherever the two differ.
- This is a first version. Help us improve it by sharing your feedback with your teacher.

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Preface

In *Linear Algebra I* you learned the structure of vector spaces: subspaces, bases, dimension, linear maps. In *Linear Algebra II* you learned to see **inside** a transformation — the determinant detecting invertibility, the eigenvalues and eigenvectors revealing the directions a matrix merely scales, and diagonalization $A = PDP^{-1}$ exposing a map's intrinsic content once you choose the right basis. Two pieces of geometry, however, were missing from both courses: there was no way to speak of the **length** of a vector, and no way to speak of the **angle** between two vectors. A vector space, as you knew it, was pure algebra without a ruler or a protractor.

This course supplies the missing instrument. Everything flows from adding a single structure — the **scalar product** (or **inner product**) — a way of multiplying two vectors to get a number. From that one number length and angle both emerge, and with them a cascade of geometry: **orthogonality** (vectors at right angles), **orthonormal bases** (the cleanest coordinate systems there are), and the **Gram–Schmidt process** that manufactures them. A vector space equipped with a scalar product is called an **inner product space** — a **Euclidean space** when it is finite-dimensional — and it is the natural home of geometry done with the tools of linear algebra.

The course then reaches two high points. The first is the **spectral theorem**: every real **symmetric** matrix is not merely diagonalizable but **orthogonally** diagonalizable, $A = PDP^T$ with P orthogonal — its eigenvectors can always be chosen mutually perpendicular. This is the cleanest and most useful diagonalization theorem in linear algebra: for symmetric matrices, the diagonalization can never fail, unlike in *Linear Algebra II*. The second high point is **orthogonal projection** and the **best-approximation theorem**: given a vector and a subspace, there is a unique closest point in the subspace, and we can compute it. The distance to that point is the geometric idea behind **least-squares regression**.

These are not abstract curiosities. The scalar product is **cosine similarity**, used everywhere in search and recommendation. The spectral theorem is the basis of **principal component analysis**. Positive-definiteness classifies the critical points found by the **Hessian** in optimization. Best approximation **is** least-squares. You are building the mathematical foundation of data science.

What you must bring, and what no chapter here will teach you, is the linear algebra of the two courses above: vector spaces, bases, dimension and linear maps from the first, and determinants, eigenvalues and diagonalization from the second. One thing more is borrowed that neither course covered — the geometry of polynomials and functions rests on the **definite integral** — so the few facts of integral calculus these examples use are recalled once, where they are first needed, and nothing else outside those two courses is assumed.

Keep one idea in view: **the scalar product turns algebra into geometry — and the first question of geometry is always “how close?”**

1. The scalar product

Vectors in $\mathbb{R}^n$ can be added and scaled, but so far there has been no way to measure one against another. The **scalar product** is the tool that does this: it takes two vectors and returns a single number, and from that number both length and angle will be reconstructed. We begin with the familiar case in the plane, extract the properties that make it useful, and then promote those properties to a **definition** that works far beyond $\mathbb{R}^n$.

1.1. A first example: the dot product in the plane

You have already met one scalar product, perhaps without the name. For two vectors $u = (u_1, u_2)$ and $v = (v_1, v_2)$ in $\mathbb{R}^2$, their **dot product** is

$$u \cdot v = u_1 v_1 + u_2 v_2.$$

Take $u = (3, 0)$ and $v = (3, 4)$. Then $u \cdot v = 9$. Two facts about this number are worth noticing immediately. First, $u \cdot u = 9 = 3^2$ is the **square of the length** of u . Second, when we compute $v \cdot v = 9 + 16 = 25 = 5^2$, we recover the length 5 of v — exactly Pythagoras. The dot product secretly **knows the geometry** of the plane. Our whole task is to isolate why, and to make the same trick work in spaces where we have no picture to fall back on.

1.2. The axioms

What properties of the dot product are doing the work? Three, and they are all the structure we need.

Definition 1. Let E be a real vector space. A **scalar product** (or **inner product**) on E is a map $\langle \cdot, \cdot \rangle : E \times E \longrightarrow \mathbb{R}$ that is

1. **symmetric**: $\langle u, v \rangle = \langle v, u \rangle$ for all u, v ;
2. **bilinear**: linear in each argument separately — for all u, v, w and scalars a, b ,

$$\langle au + bv, w \rangle = a\langle u, w \rangle + b\langle v, w \rangle$$

(and likewise in the second argument, which then follows by symmetry);

3. **positive definite**: $\langle u, u \rangle \geq 0$ for all u , with $\langle u, u \rangle = 0$ **only** when $u = 0$.

A real vector space equipped with a scalar product is an **inner product space**; a finite-dimensional one is a **Euclidean space**.

The three axioms are not arbitrary. **Symmetry** says the order does not matter. **Bilinearity** says the product interacts with linear combinations exactly the way multiplication should — it is the property that lets us compute. **Positive definiteness** is the crucial one: it guarantees that $\langle u, u \rangle$ is a genuine “squared length”, never negative, and zero only for the zero vector. Drop it and lengths could be imaginary or vanish on non-zero vectors.

Pitfall. All three axioms are required, and positive **definiteness** is stronger than mere positivity. A form can satisfy $\langle u, u \rangle \geq 0$ for all u yet vanish on some non-zero vector — for example

$\langle u, v \rangle = u_1 v_1$ on $\mathbb{R}^2$ gives $\langle (0, 5), (0, 5) \rangle = 0$ with $(0, 5) \neq 0$. Such a form is **positive semi-definite** but not a scalar product: it cannot tell the vector $(0, 5)$ apart from 0.

1.3. The standard scalar product on $\mathbb{R}^n$

The dot product of the opening example generalizes verbatim to any dimension.

Definition 2. The **standard** (or **canonical**) scalar product on $\mathbb{R}^n$ is

$$\langle u, v \rangle = u^\top v = \sum_{i=1}^n u_i v_i = u_1 v_1 + u_2 v_2 + \dots + u_n v_n.$$

That this satisfies the three axioms is a direct check: symmetry is the commutativity of real multiplication; bilinearity is the distributive law; and $\langle u, u \rangle = \sum u_i^2$ is a sum of squares, hence ≥ 0 and zero only when every $u_i = 0$. This is the scalar product unless another is stated, but — and this is the key new idea of the chapter — it is **not the only one**.

1.4. Other Euclidean spaces: polynomials and matrices

The power of the axiomatic definition is that it applies to vector spaces whose elements are not lists of numbers at all. Two examples matter most. The first uses a little integral calculus, so we recall the few facts it needs before building it.

Remark (calculus recalled). The polynomial and function spaces of this course borrow a small amount of integral calculus, which neither Linear Algebra course assumed. Only these facts are used, and every integral in the book can be checked from them. The **definite integral** $\int_a^b f(x) dx$ of a continuous function f over an interval $[a, b]$ is a real number, and it is **linear**:

$$\int_a^b (\alpha f(x) + \beta g(x)) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx.$$

On a power of x it is given by the **power rule**

$$\int_a^b x^n dx = \frac{b^{n+1} - a^{n+1}}{n+1} \quad (n \text{ an integer } \geq 0),$$

and on the trigonometric functions by

$$\int_a^b \cos x dx = \sin b - \sin a, \quad \int_a^b \sin x dx = \cos a - \cos b.$$

Finally, a continuous function that is ≥ 0 on $[a, b]$ and whose integral over $[a, b]$ is 0 is **identically zero** there: a point where it were positive would, by continuity, contribute a positive area. This last fact is what makes the integral positive definite below.

Example — an inner product on polynomials. On the space $\mathbb{R}[X]_{\leq n}$ of polynomials of degree at most n , define

$$\langle P, Q \rangle = \int_0^1 P(x)Q(x) dx.$$

This is symmetric (multiplication of functions commutes) and bilinear (the integral is linear). For positive definiteness, $\langle P, P \rangle = \int_0^1 P(x)^2 dx \geq 0$, and a continuous non-negative function with integral zero must be identically zero — so $\langle P, P \rangle = 0$ forces $P = 0$. Thus polynomials acquire **lengths and angles**: we can ask whether two polynomials are “orthogonal”, a question with no meaning until a scalar product is fixed.

Example — an inner product on matrices. On the space $\mathcal{M}_n(\mathbb{R})$ of square matrices, the **Frobenius** inner product is

$$\langle A, B \rangle = \text{Tr}(A^T B) = \sum_{i,j} a_{ij} b_{ij}.$$

The last equality shows it is **literally** the standard scalar product on $\mathbb{R}^{n^2}$ — flatten each matrix into a long list of its entries and take the dot product. So $\langle A, A \rangle = \sum_{i,j} a_{ij}^2$ is the sum of squared entries, manifestly positive definite.

Remark. One vector space can carry **many** different scalar products. On $\mathbb{R}^2$, besides the standard one, $\langle u, v \rangle = 2u_1v_1 + 3u_2v_2$ is also a scalar product (a **weighted** dot product). Each choice bends the geometry: lengths and angles depend on which scalar product is in force. When we speak of “the angle between two vectors” we always mean **relative to a chosen scalar product**.

2. Norm, Cauchy–Schwarz, and angle

With a scalar product in hand we can finally **measure**. The squared length of a vector is $\langle u, u \rangle$; taking a square root gives its length, or **norm**. The deeper miracle is that the scalar product also encodes **angle** — and the bridge between the two is a single inequality, the most important in the course.

2.1. The norm

In the plane the length of $(3, 4)$ is $\sqrt{3^2 + 4^2} = 5$, which is exactly $\sqrt{\langle (3, 4), (3, 4) \rangle}$ under the standard product. Taking the square root of the scalar product of a vector with itself is what turns a squared length into a length, and the same recipe works in any inner product space.

Definition 3. In an inner product space, the **norm** (or **length**) of a vector u is

$$\|u\| = \sqrt{\langle u, u \rangle}.$$

A vector with $\|u\| = 1$ is a **unit vector**. For any $u \neq 0$, the vector $u/\|u\|$ is a unit vector in the same direction; replacing u by it is called **normalizing**.

The square root is well-defined precisely because positive definiteness makes $\langle u, u \rangle \geq 0$. In the standard product on $\mathbb{R}^n$ this is $\|u\| = \sqrt{u_1^2 + \dots + u_n^2}$, the ordinary Euclidean length. In the polynomial space it is $\|P\| = \sqrt{\int_0^1 P(x)^2 dx}$, a perfectly good “length” of a polynomial.

Proposition 1 (Basic properties of the norm). For all u and scalars λ :

- $\|u\| \geq 0$, with $\|u\| = 0$ if and only if $u = 0$;
- $\|\lambda u\| = |\lambda| \|u\|$ (absolute **homogeneity**).

2.2. The Cauchy–Schwarz inequality

This is the central inequality of the course. It bounds the scalar product of two vectors by the product of their lengths — and it is what allows angle to be **defined**.

Try it on $u = (1, 2, 2)$ and $v = (2, 2, 1)$ in $\mathbb{R}^3$: here $\langle u, v \rangle = 2 + 4 + 2 = 8$, while $\|u\| = \|v\| = \sqrt{9} = 3$, so $\|u\| \|v\| = 9$. The scalar product 8 stays under the product of lengths 9 — and only just, the ratio $8/9 \approx 0.89$ being close to 1 because u and v point in nearly the same direction. The theorem says this is no accident: the scalar product can **never** exceed the product of the lengths, in any inner product space.

Theorem 1 (Cauchy–Schwarz inequality). For all vectors u, v in an inner product space,

$$|\langle u, v \rangle| \leq \|u\| \|v\|,$$

with **equality** if and only if u and v are **collinear** (one is a scalar multiple of the other).

Proof. For every real t , build the number

$$P(t) = \|u + tv\|^2,$$

which is ≥ 0 by positive definiteness. Expanding it by bilinearity and symmetry,

$$P(t) = \langle u + tv, u + tv \rangle = \|u\|^2 + 2t\langle u, v \rangle + t^2\|v\|^2.$$

Dispose of the degenerate case first: if $v = 0$ then both sides of the claimed inequality are 0 and there is nothing to prove, so assume $v \neq 0$. Then $\|v\|^2 > 0$, so P is a genuine quadratic in t with positive leading coefficient that is never negative; its discriminant is therefore ≤ 0 :

$$(2\langle u, v \rangle)^2 - 4\|v\|^2\|u\|^2 \leq 0, \quad \text{that is} \quad \langle u, v \rangle^2 \leq \|u\|^2\|v\|^2.$$

Taking square roots gives $|\langle u, v \rangle| \leq \|u\| \|v\|$.

For the equality case, suppose $|\langle u, v \rangle| = \|u\| \|v\|$ with $v \neq 0$. Then the discriminant is exactly 0, so the quadratic P has a real double root t_0 , at which $P(t_0) = \|u + t_0v\|^2 = 0$; positive definiteness forces $u + t_0v = 0$, so $u = -t_0v$ and u, v are collinear. Conversely, if $u = \lambda v$ then $|\langle u, v \rangle| = |\lambda|\|v\|^2 = \|u\| \|v\|$, so collinearity gives equality. (When $v = 0$ the vectors are trivially collinear and both sides are 0.) $\square$

A first consequence of Cauchy–Schwarz is that norms add the way lengths should.

Proposition 2 (Triangle inequality). $\|u + v\| \leq \|u\| + \|v\|$ for all u, v .

Proof. Expanding and applying Cauchy–Schwarz,

$$\|u + v\|^2 = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2.$$

Take square roots. $\square$

2.3. Angle and cosine similarity

Because Cauchy–Schwarz guarantees that the ratio $\langle u, v \rangle / (\|u\| \|v\|)$ always lies in $[-1, 1]$, it is exactly the cosine of an angle.

Definition 4. The **angle** $\theta \in [0, \pi]$ between two non-zero vectors u, v is defined by

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}.$$

In data science this quantity is called the **cosine similarity** of u and v .

The definition is legitimate **only** because Cauchy–Schwarz keeps the right-hand side within $[-1, 1]$, the range of cosine. This is the payoff of the inequality: it is what makes “angle” meaningful in spaces — polynomials, matrices, documents-as-word-count-vectors — where no protractor could ever reach.

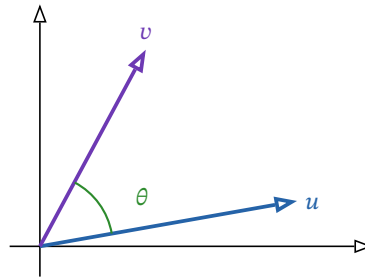


Figure 1: The angle θ between u and v is recovered from the scalar product alone: $\cos \theta = \langle u, v \rangle / (\|u\| \|v\|)$. Cauchy–Schwarz guarantees this ratio lies in $[-1, 1]$, so the angle always exists. Parallel vectors give $\cos \theta = \pm 1$; perpendicular ones give $\cos \theta = 0$.

Pitfall. Cosine similarity measures **direction**, not magnitude. The vectors $(1, 1)$ and $(1000, 1000)$ have cosine similarity 1 — they are perfectly “similar” though hugely different in length. This is exactly why it suits comparing documents of very different sizes, but it also means a near-1 cosine says nothing about how **large** the vectors are.

2.4. A wider world of norms

Everything so far measured length with the **induced** norm $\|u\| = \sqrt{\langle u, u \rangle}$, born from a scalar product. But “length” is a more general idea: any function assigning a sensible size to vectors deserves the name, whether or not a scalar product stands behind it. Isolating the three properties that make the induced norm work gives the general definition.

Definition 5. A **norm** on a real vector space E is a map $N : E \rightarrow \mathbb{R}$, written $N(u) = \|u\|$, such that for all u, v and scalars λ :

1. **positive definiteness:** $\|u\| \geq 0$, with $\|u\| = 0$ only for $u = 0$;
2. **homogeneity:** $\|\lambda u\| = |\lambda| \|u\|$;
3. **triangle inequality:** $\|u + v\| \leq \|u\| + \|v\|$.

The induced norm satisfies all three — the first two are the norm’s basic properties recorded earlier, the third is the triangle inequality proved above from Cauchy–Schwarz. But on $\mathbb{R}^n$ there are useful norms of a different origin, not induced by any scalar product.

Example — three norms on the same space. For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, three standard norms are

$$\|x\|_1 = \sum_{i=1}^n |x_i|, \quad \|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}, \quad \|x\|_\infty = \max_i |x_i|.$$

The middle one, $\|x\|_2$, is the familiar Euclidean (induced) norm. The **taxicab** norm $\|x\|_1$ adds the coordinate distances — total travel on a grid of streets. The **maximum** norm $\|x\|_\infty$ reports the single largest coordinate. Each obeys the three axioms, so each is a legitimate way to measure length; on $x = (3, -4)$ they read 7, 5, and 4 respectively.

A norm is pictured by its **unit ball**

$$\{x : \|x\| \leq 1\}$$

— the set of vectors of length at most 1. Its shape is the signature of the norm.

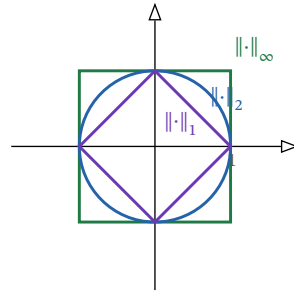


Figure 2: The unit balls of the three norms on $\mathbb{R}^2$, nested by the inequalities $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$: a larger ball means an **easier** length to satisfy. The taxicab norm $\|\cdot\|_1$ gives a **diamond** (purple), the Euclidean $\|\cdot\|_2$ the round **disc** (blue), the maximum $\|\cdot\|_\infty$ a **square** (green). Only the round one comes from a scalar product — a corner is a place no inner product can produce.

Remark (which norms come from a scalar product). Not every norm is induced by a scalar product. A norm that must obey the **parallelogram law** $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$, which follows by expanding both norms with the scalar product. Test $\|\cdot\|_1$ on $u = (1, 0)$, $v = (0, 1)$: the left side is $2^2 + 2^2 = 8$ (since $\|u \pm v\|_1 = 2$), the right side $2 + 2 = 4$. The law fails, so $\|\cdot\|_1$ comes from **no** inner product — and likewise $\|\cdot\|_\infty$. Their unit balls have corners; the induced norm's is always a smooth ellipsoid.

Different norms rank vectors differently, yet on a finite-dimensional space they never disagree **wildly**. This is the last basic fact about norms.

Theorem 2 (Equivalence of norms in finite dimension). On a finite-dimensional real vector space, any two norms N and N' are **equivalent**: there are constants $c, C > 0$ with

$$cN(x) \leq N'(x) \leq CN(x) \quad \text{for all } x.$$

Concretely, on $\mathbb{R}^n$,

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2 \leq n \|x\|_\infty.$$

Proof. The chain of concrete inequalities is elementary. Since each $|x_i| \leq \|x\|_\infty$, the largest square is at most the sum of squares: $\|x\|_\infty \leq \|x\|_2$. Squaring $\|x\|_1$ expands to $\sum_i x_i^2$ plus non-negative cross terms $2 \sum_{i < j} |x_i| |x_j|$, so $\|x\|_2 \leq \|x\|_1$. Cauchy-Schwarz applied to the vectors $(|x_i|)_i$ and $(1, \dots, 1)$ gives $\|x\|_1 = \sum_i |x_i| \cdot 1 \leq \|x\|_2 \sqrt{n}$. Finally $\|x\|_2 \leq \sqrt{n} \|x\|_\infty$ because each of the n squares is at most $\|x\|_\infty^2$. The general statement — that **every** pair of norms is related this way in finite dimension — rests on the compactness of the unit sphere and is left to a later analysis course. $\square$

Remark. Equivalence has a sharp consequence: all norms on a finite-dimensional space induce the **same** notion of “close”. A sequence converging in one norm converges in every other, to the same limit, since being small in N' forces being small in N and back. The **geometry** (which vectors are longer) depends on the norm; the **topology** (which vectors are near) does not. This

is special to finite dimension — in infinite-dimensional spaces of functions, different norms can genuinely disagree about convergence.

3. Orthogonality and orthonormal bases

The single most useful angle is the right angle. When $\cos \theta = 0$ — that is, when $\langle u, v \rangle = 0$ — the vectors are **orthogonal**, and orthogonality turns out to be the organizing principle of the entire course. Families of mutually orthogonal vectors are automatically independent, and bases built from unit-length orthogonal vectors make every computation effortless.

3.1. Orthogonal and orthonormal families

The two standard axes $(1, 0)$ and $(0, 1)$ of the plane are perpendicular and each of length one: their scalar product is 0 and each has norm 1. That pair of features — mutual perpendicularity and unit length — is what the following definition names in general.

Definition 6. Two vectors u, v are **orthogonal**, written $u \perp v$, if $\langle u, v \rangle = 0$. A family $(e_1, \dots, e_k)$ is

- **orthogonal** if its vectors are pairwise orthogonal: $\langle e_i, e_j \rangle = 0$ whenever $i \neq j$;
- **orthonormal** if it is orthogonal and every vector is a unit vector: $\langle e_i, e_j \rangle = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$.

The standard basis $(e_1, \dots, e_n)$ of $\mathbb{R}^n$ is the prototype orthonormal family. Orthonormality is the cleanest possible relationship a family can have: it says the vectors are mutually perpendicular **and** individually of unit length.

Example — an orthonormal pair. In $\mathbb{R}^2$ the vectors $e_1 = \frac{1}{\sqrt{2}}(1, 1)$ and $e_2 = \frac{1}{\sqrt{2}}(1, -1)$ satisfy $\langle e_1, e_2 \rangle = \frac{1}{2}(1 - 1) = 0$ and $\|e_1\| = \|e_2\| = 1$. They are the standard axes rotated by 45° — a different orthonormal basis of the same plane.

3.2. Orthogonality forces independence

Here is the first structural payoff, and it is a strong one.

Proposition 3 (orthogonal families are independent). Any orthogonal family of **non-zero** vectors is linearly independent.

Proof. Suppose $(e_1, \dots, e_k)$ is orthogonal with each $e_i \neq 0$, and $a_1 e_1 + \dots + a_k e_k = 0$. Take the scalar product of both sides with a fixed e_j . By bilinearity and orthogonality, every term $a_i \langle e_i, e_j \rangle$ vanishes except $i = j$, leaving

$$a_j \langle e_j, e_j \rangle = 0.$$

Since $e_j \neq 0$, positive definiteness gives $\langle e_j, e_j \rangle > 0$, forcing $a_j = 0$. As j was arbitrary, all coefficients are zero. $\square$

Remark. This is genuinely useful: checking independence normally means row-reducing a matrix, but for an orthogonal family no computation is needed — independence is automatic.

In particular, n non-zero orthogonal vectors in an n -dimensional space form a basis with no further work.

Pitfall. The vectors must be **non-zero**. The zero vector is orthogonal to everything (including itself), so a family containing 0 is “orthogonal” in the literal sense yet certainly not independent. Orthonormal families avoid this automatically, since unit vectors are never zero.

3.3. Coordinates in an orthonormal basis are read off directly

The reason orthonormal bases are so valuable is that coordinates, which normally require solving a linear system, are read off by **single scalar products**.

Proposition 4 (Fourier coefficients). If $(e_1, \dots, e_n)$ is an orthonormal basis of E , then every $v \in E$ expands as

$$v = \sum_{i=1}^n \langle v, e_i \rangle e_i.$$

The coordinate of v along e_i is simply $\langle v, e_i \rangle$. Moreover (**Pythagoras / Parseval**) $\|v\|^2 = \sum_{i=1}^n \langle v, e_i \rangle^2$.

Proof. Write $v = \sum_j a_j e_j$ in the basis. Take $\langle \cdot, e_i \rangle$: orthonormality kills every term but $j = i$, giving $\langle v, e_i \rangle = a_i$. For the norm, expand $\langle v, v \rangle = \sum_{i,j} a_i a_j \langle e_i, e_j \rangle = \sum_i a_i^2$. $\square$

Example — reading off coordinates. With the rotated basis $e_1 = \frac{1}{\sqrt{2}}(1, 1)$, $e_2 = \frac{1}{\sqrt{2}}(1, -1)$ and $v = (3, 1)$:

$$\langle v, e_1 \rangle = \frac{3+1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}, \quad \langle v, e_2 \rangle = \frac{3-1}{\sqrt{2}} = \sqrt{2}.$$

So $v = 2\sqrt{2}e_1 + \sqrt{2}e_2$, and indeed $\|v\|^2 = 8 + 2 = 10 = 3^2 + 1^2$. ✓ No linear system had to be solved — just two dot products.

This single property is why the rest of the course works so hard to **build** orthonormal bases. The tool that builds them is next.

4. The Gram–Schmidt process

Orthonormal bases are wonderful, but a basis handed to you is rarely orthonormal. The **Gram–Schmidt process** is the algorithm that repairs this: it takes **any** basis and manufactures an orthonormal one occupying exactly the same space, one vector at a time.

4.1. The idea: subtract off what overlaps

The core move is to take a new vector and **remove its component along the directions already handled**, leaving a remainder orthogonal to all of them. The component of v along a unit vector e is $\langle v, e \rangle e$; subtracting it leaves $v - \langle v, e \rangle e$, which is orthogonal to e by construction.

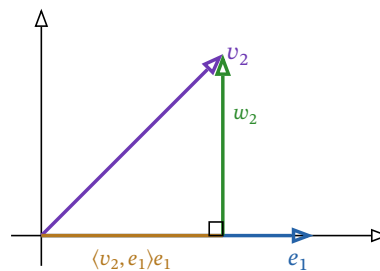


Figure 3: One Gram–Schmidt step. From v_2 subtract its component along the already-fixed direction e_1 ; the remainder $w_2 = v_2 - \langle v_2, e_1 \rangle e_1$ is orthogonal to e_1 (right angle marked). Normalizing w_2 gives e_2 . Each new vector is purged of every direction already in the basis.

4.2. The algorithm

Theorem 3 (Gram–Schmidt process). Let $(v_1, \dots, v_k)$ be a linearly independent family in an inner product space. Define recursively

$$w_1 = v_1, \quad w_j = v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} w_i \quad (j = 2, \dots, k),$$

and set $e_j = w_j / \|w_j\|$. Then $(e_1, \dots, e_k)$ is **orthonormal** and, for every j ,

$$\text{Span}(e_1, \dots, e_j) = \text{Span}(v_1, \dots, v_j).$$

The displayed **span-preservation** property is as important as the orthonormality: Gram–Schmidt does not merely produce **some** orthonormal family, it produces one that fills the **same nested subspaces** as the original. The first j outputs span exactly what the first j inputs did.

Proof. By induction each w_j is, by construction, the part of v_j orthogonal to all earlier w_i : taking $\langle \cdot, w_m \rangle$ for $m < j$ and using that the earlier w_i are already mutually orthogonal, every term cancels except the one that subtracts off the overlap, giving $\langle w_j, w_m \rangle = 0$. Since each w_j differs from v_j only by a combination of earlier vectors, the spans coincide at each stage. Normalizing rescales without changing direction or span. Independence of the v_i guarantees no w_j is zero, so the division is legitimate. $\square$

Example — Gram–Schmidt in $\mathbb{R}^3$. Orthonormalize $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$.

Step 1. $w_1 = (1, 1, 0)$, $\|w_1\| = \sqrt{2}$, so $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.

Step 2. $\langle v_2, w_1 \rangle = 1$, $\langle w_1, w_1 \rangle = 2$, so

$$w_2 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right),$$

with $\|w_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$, giving $e_2 = \sqrt{\frac{2}{3}}\left(\frac{1}{2}, -\frac{1}{2}, 1\right) = \frac{1}{\sqrt{6}}(1, -1, 2)$.

Step 3. $\langle v_3, w_1 \rangle = 1$ and $\langle v_3, w_2 \rangle = -\frac{1}{2} + 1 = \frac{1}{2}$:

$$w_3 = (0, 1, 1) - \frac{1}{2}(1, 1, 0) - \frac{\frac{1}{2}}{\frac{3}{2}}\left(\frac{1}{2}, -\frac{1}{2}, 1\right) = (0, 1, 1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) - \left(\frac{1}{6}, -\frac{1}{6}, \frac{1}{3}\right).$$

This is $w_3 = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$, with $\|w_3\| = \frac{2}{\sqrt{3}}$, so $e_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)$. The three e_i are orthonormal — check, e.g., $\langle e_1, e_2 \rangle = \frac{1-1+0}{\sqrt{12}} = 0$. ✓

Pitfall. Two slips are common. First, when not normalizing as you go, you must divide by $\langle w_i, w_i \rangle$ — **not** by $\langle v_i, v_i \rangle$ — in the projection coefficients; project onto the **already-orthogonalized** w_i , never the raw v_i . Second, the process needs the input to be **independent**; if the v_j are dependent, some w_j comes out 0 and the normalization divides by zero — the signal that v_j was already in the span of its predecessors.

4.3. Application to function and polynomial spaces

Because Gram–Schmidt uses only the scalar product, it runs unchanged in spaces of polynomials or functions, where it produces families of **orthogonal polynomials** and **orthogonal functions**.

Example — Legendre polynomials. Apply Gram–Schmidt to $1, x, x^2$ on $\mathbb{R}[X]_{\leq 2}$ with $\langle P, Q \rangle = \int_{-1}^1 PQ dx$. Note the interval here is $[-1, 1]$, not the $[0, 1]$ used elsewhere in this book: a different interval is a different scalar product, and the Legendre polynomials are the ones tied to the **symmetric** interval $[-1, 1]$. Start with $w_1 = 1$. Then $\langle x, 1 \rangle = \int_{-1}^1 x dx = 0$, so x is **already** orthogonal to 1: $w_2 = x$. For x^2 , $\langle x^2, 1 \rangle = \frac{2}{3}$ and $\langle 1, 1 \rangle = 2$, while $\langle x^2, x \rangle = 0$, so

$$w_3 = x^2 - \frac{\frac{2}{3}}{2} \cdot 1 = x^2 - \frac{1}{3}.$$

The outputs $1, x, x^2 - \frac{1}{3}$ are the first **Legendre polynomials** (up to scaling) — the orthogonal basis underlying Gaussian quadrature and many approximation schemes.

The next example leaves polynomials behind entirely, orthonormalizing genuine **non-polynomial functions**.

Example — the Fourier basis. On the space of continuous functions on $[0, 2\pi]$ with

$$\langle f, g \rangle = \int_0^{2\pi} f(x)g(x) dx,$$

orthonormalize the three functions $1, \cos x, \sin x$ — none of them a polynomial. Their pairwise scalar products all vanish:

$$\langle 1, \cos x \rangle = \int_0^{2\pi} \cos x dx = 0, \quad \langle 1, \sin x \rangle = 0, \quad \langle \cos x, \sin x \rangle = 0,$$

where the last two use the identity $\cos x \sin x = (\sin 2x)/2$, whose integral over the full period $[0, 2\pi]$ is 0. So the family is **already orthogonal** — Gram–Schmidt subtracts nothing and only normalizes. Since $\langle 1, 1 \rangle = 2\pi$ and, using $\cos^2 x = (1 + \cos 2x)/2$ and $\sin^2 x = (1 - \cos 2x)/2$ (again the $\cos 2x$ term integrates to 0 over the period), $\langle \cos x, \cos x \rangle = \langle \sin x, \sin x \rangle = \pi$, the orthonormal family is

$$\frac{1}{\sqrt{2\pi}}, \quad \frac{\cos x}{\sqrt{\pi}}, \quad \frac{\sin x}{\sqrt{\pi}}.$$

These are the first vectors of the **Fourier basis**, the function-space analogue of an orthonormal basis of $\mathbb{R}^n$. Reading a function's coordinates in it by single scalar products $\langle f, e_i \rangle$ — exactly the orthonormal-basis coordinate rule, now applied to functions — is what computes its **Fourier coefficients**. Their mutual orthogonality, checked above, is precisely why that expansion works.

5. Orthogonal matrices

Gram–Schmidt produces orthonormal bases; we now study the matrices that **map one orthonormal basis to another**. These are the **orthogonal matrices** — the matrices that preserve all lengths and angles — the rotations and the orientation-reversing rigid motions of space that fix the origin.

5.1. Definition and the column criterion

Look at the plane rotation $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$: its two columns each have length one and are perpendicular to each other. Matrices whose columns behave this way are exactly those that preserve every length and angle, and they earn their own name.

Definition 7. A matrix $P \in \mathcal{M}_n(\mathbb{R})$ is **orthogonal** if

$$P^T P = I, \quad \text{equivalently} \quad P^T = P^{-1}.$$

The condition has a vivid reading in terms of columns. The (i, j) entry of $P^T P$ is the dot product of column i with column j of P ; demanding $P^T P = I$ says those dot products are 1 on the diagonal and 0 off it.

Proposition 5 (Orthogonal = orthonormal columns). P is orthogonal if and only if its columns form an **orthonormal basis** of $\mathbb{R}^n$. (Equivalently, since $PP^T = I$ too, its rows do as well.)

Example — a rotation matrix. The rotation by angle θ ,

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

has columns $(\cos \theta, \sin \theta)$ and $(-\sin \theta, \cos \theta)$: each has norm $\sqrt{\cos^2 + \sin^2} = 1$, and their dot product is $-\cos \theta \sin \theta + \sin \theta \cos \theta = 0$. So R is orthogonal for every θ .

5.2. Orthogonal maps preserve the geometry

The defining property is exactly preservation of the scalar product, hence of all lengths and angles.

Proposition 6 (Isometry). If P is orthogonal then for all u, v ,

$$\langle Pu, Pv \rangle = \langle u, v \rangle, \quad \text{and in particular} \quad \|Pu\| = \|u\|.$$

Proof. $\langle Pu, Pv \rangle = (Pu)^T (Pv) = u^T P^T P v = u^T I v = \langle u, v \rangle$. Setting $v = u$ gives the norm statement. $\square$

So an orthogonal matrix is a **rigid motion fixing the origin**: it can turn space or flip it, but it never stretches, shrinks, or shears. This is why orthonormal bases are interchangeable — the change-of-basis matrix between any two of them is orthogonal.

5.3. Rotations versus reflections: the determinant criterion

Taking determinants of $P^T P = I$ pins down a sharp dichotomy.

Proposition 7. If P is orthogonal then $\det P = \pm 1$.

Proof. $\det(P^T P) = \det(P^T) \det(P) = (\det P)^2$, and this equals $\det I = 1$, so $\det P = \pm 1$. $\square$

Definition 8. An orthogonal matrix P with $\det P = +1$ is a **rotation**: it preserves orientation. One with $\det P = -1$ is **orientation-reversing**: it reverses orientation.

Example — classifying in the plane. The matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is orthogonal ($P^T P = I$) with $\det P = -1$, so it is orientation-reversing; concretely it sends (x, y) to $(x, -y)$, the **reflection** across the x -axis. The rotation $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ has $\det = +1$ and turns the plane through the angle θ .

In the plane every orientation-reversing orthogonal matrix is a genuine **reflection** across a line through the origin — the mirror picture is exact there, and this is the case drawn below. From dimension three on, orientation-reversing maps are more varied: some are true reflections, fixing a whole hyperplane and flipping only the perpendicular direction, while others — the **rotoreflections** — turn space **and** flip it, fixing no direction at all. What the determinant alone certifies is the **orientation-reversing** character; the name **reflection** is kept for the maps that really do fix a hyperplane.

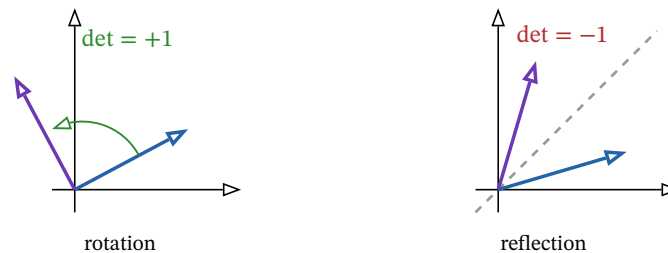


Figure 4: The two kinds of orthogonal map in the plane, both preserving lengths and angles. A **rotation** ($\det = +1$, left) keeps orientation: the ordered pair (blue, purple) stays counter-clockwise. In the plane an orientation-reversing map ($\det = -1$, right) is a **reflection**, flipping every vector across a mirror line (dashed). In higher dimensions $\det = -1$ still means orientation-reversing but no longer means a plain mirror.

Pitfall. $\det P = -1$ does **not** by itself make P orthogonal — orthogonality is the condition $P^T P = I$, and the determinant test only **classifies** a matrix already known to be orthogonal. Check $P^T P = I$ first; then the sign of the determinant tells you rotation versus orientation-reversing.

Remark. Orthogonal matrices form a group: products and inverses of orthogonal matrices are orthogonal (if $P^T P = I$ and $Q^T Q = I$ then $(PQ)^T (PQ) = Q^T P^T P Q = I$). The determinant is multiplicative, so composing two orientation-preserving maps, or two orientation-reversing ones, yields an orientation-preserving map (a rotation), while composing one of each reverses orientation: the signs simply multiply, $(+1)(+1) = (-1)(-1) = +1$ and $(+1)(-1) = -1$.

6. Symmetric matrices and the spectral theorem

We now apply the geometry of orthogonality to the eigenvalue theory of *Linear Algebra II*. The result is the most important theorem of the course. For a special but very common class of matrices — the **symmetric** ones — diagonalization always succeeds, the eigenvalues are always real, and the eigenvectors can always be chosen mutually **orthogonal**. None of the difficulties of *Linear Algebra II* (complex roots, too few independent eigenvectors) can occur.

6.1. Symmetric matrices and self-adjointness

Definition 9. A matrix $A \in \mathcal{M}_n(\mathbb{R})$ is **symmetric** if $A^\top = A$, i.e. $a_{ij} = a_{ji}$.

The reason symmetry interacts so well with the scalar product is the following identity, which says A can be moved from one side of an inner product to the other.

Proposition 8 (Self-adjointness). A is symmetric if and only if $\langle Au, v \rangle = \langle u, Av \rangle$ for all $u, v \in \mathbb{R}^n$.

Proof. $\langle Au, v \rangle = (Au)^\top v = u^\top A^\top v$ and $\langle u, Av \rangle = u^\top Av$. These agree for all u, v exactly when $A^\top = A$. $\square$

6.2. Two structural facts

Symmetry forces two remarkable properties before we even reach diagonalization.

Remark (a brief word on complex vectors). The proof just below needs to reason about a **possibly complex** eigenvalue, so we borrow three facts. For a complex number $z = a + bi$, its **conjugate** is $\bar{z} = a - bi$, and $\bar{z}z = a^2 + b^2 = |z|^2 \geq 0$ is a real number, zero only when $z = 0$. Conjugation respects sums and products: $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{zw} = \bar{z}\bar{w}$. For a vector $v \in \mathbb{C}^n$, $\bar{v}$ conjugates each entry, so $\bar{v}^\top v = \sum_i \bar{v}_i v_i = \sum_i |v_i|^2 \geq 0$ is real and strictly positive unless $v = 0$. Finally, a matrix with **real** entries is its own conjugate, $\bar{A} = A$. These few facts are all the proof uses; no further complex analysis is needed.

Proposition 9 (Real eigenvalues). Every eigenvalue of a real symmetric matrix is **real**.

Proof. We have to be careful: a real matrix can **a priori** have complex eigenvalues (its characteristic polynomial is a real polynomial, whose roots may be complex). So we temporarily allow λ and the eigenvector v to be complex, and show that in fact λ must be real. Write $\bar{v}$ for the entrywise complex conjugate of v , and consider the single number $\bar{v}^\top Av$.

On one hand, using $Av = \lambda v$,

$$\bar{v}^T A v = \lambda \bar{v}^T v.$$

On the other hand, $\bar{v}^T A v$ is a 1×1 quantity, so it equals its own transpose; transposing and conjugating (and using that A has real symmetric entries, $\bar{A}^T = A$) turns it into $\bar{\lambda} \bar{v}^T v$. Comparing, $(\lambda - \bar{\lambda}) \bar{v}^T v = 0$. Now $\bar{v}^T v = \sum_i \bar{v}_i v_i = \sum_i |v_i|^2$ is a strictly positive **real** number (as $v \neq 0$), so $\lambda = \bar{\lambda}$: the eigenvalue is real. $\square$

Proposition 10 (Orthogonal eigenspaces). Eigenvectors of a symmetric matrix for **distinct** eigenvalues are orthogonal.

Proof. Let $Au = \lambda u$ and $Av = \mu v$ with $\lambda \neq \mu$. Using self-adjointness,

$$\lambda \langle u, v \rangle = \langle Au, v \rangle = \langle u, Av \rangle = \mu \langle u, v \rangle,$$

so $(\lambda - \mu) \langle u, v \rangle = 0$. As $\lambda \neq \mu$, we conclude $\langle u, v \rangle = 0$. $\square$

These two facts are important, but on their own they do **not** yet give the spectral theorem, and it is worth being honest about the gap. Real eigenvalues tell us the characteristic polynomial has all its roots in $\mathbb{R}$ — but a polynomial splitting over $\mathbb{R}$ does not by itself make a matrix diagonalizable (the matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has the single real eigenvalue 1 and is **not** diagonalizable). And orthogonality of eigenvectors from **distinct** eigenvalues says nothing about a **repeated** eigenvalue: we still need to know that a repeated eigenvalue supplies as many independent eigenvectors as its multiplicity — that there is never a shortfall. That missing guarantee is exactly the content of the spectral theorem, and it needs its own proof, given below.

6.3. The spectral theorem

Theorem 4 (Spectral theorem). Every real symmetric matrix $A \in \mathcal{M}_n(\mathbb{R})$ is **orthogonally diagonalizable**: there exist an **orthogonal** matrix P (so $P^T = P^{-1}$) and a real diagonal matrix D with

$$A = PDP^T.$$

The columns of P form an orthonormal basis of eigenvectors of A , and the diagonal of D holds the corresponding (real) eigenvalues. Conversely, any matrix of the form PDP^T with P orthogonal and D diagonal is symmetric.

Proof. We prove the main direction by induction on the size n , building an orthonormal basis of eigenvectors one vector at a time. The key geometric fact is a small lemma: if e is an eigenvector of a symmetric A , then the subspace $e^\perp$ of all vectors orthogonal to e is **sent into itself** by A . Indeed, for $w \perp e$, self-adjointness gives

$$\langle Aw, e \rangle = \langle w, Ae \rangle = \langle w, \lambda e \rangle = \lambda \langle w, e \rangle = 0,$$

so $Aw \perp e$ as well.

For $n = 1$ every 1×1 matrix is already diagonal. Assume the theorem for size $n - 1$. Given a real symmetric A of size n , its characteristic polynomial has a root, which is **real** by the real-eigenvalue proposition; pick a corresponding **unit** eigenvector e_1 (so $Ae_1 = \lambda_1 e_1$). By the lemma, A maps the

$(n-1)$ -dimensional subspace $F = e_1^\perp$ into itself, and the restriction of A to F is again self-adjoint. Expressed in an orthonormal basis of F , this restriction is a symmetric $(n-1) \times (n-1)$ matrix, so by the induction hypothesis it has an **orthonormal basis of eigenvectors** $e_2, \dots, e_n$ of F — which are also eigenvectors of A . Together with e_1 they form an orthonormal basis of $\mathbb{R}^n$ made entirely of eigenvectors of A . Placing them as the columns of P makes P orthogonal and $A = PDP^\top$, with the eigenvalues on the diagonal of D . $\square$

Notice that this argument **never mentions multiplicities**: the induction hands us exactly n orthonormal eigenvectors automatically, so the “shortfall” that can wreck an ordinary diagonalization simply cannot arise. This is the guarantee the previous section said was missing.

The converse is the easy half and worth seeing: if $A = PDP^\top$ then $A^\top = (PDP^\top)^\top = PD^\top P^\top = PDP^\top = A$, since D is diagonal. So **orthogonal diagonalizability and symmetry are the same property** — symmetric matrices are exactly those a rotation/reflection turns into pure scalings.

Pitfall. Note the $P^\top$ in place of the P^{-1} of ordinary diagonalization. Because P is orthogonal the two coincide, so $A = PDP^\top$ — but the point is that for a symmetric matrix you may **choose** P orthogonal, which is a far stronger and cleaner statement than mere diagonalizability. Inverting P is then free: just transpose.

6.4. Geometric interpretation: stretching along the eigen-axes

The decomposition $A = PDP^\top$ is more than an algebraic factorization: it says exactly **what a symmetric matrix does to space**. Applying A to a vector is three moves performed in order, read right to left in $PDP^\top$:

$$Ax = \underbrace{P}_{\text{rotate back}} \underbrace{D}_{\text{stretch axis } i \text{ by } \lambda_i} \underbrace{P^\top}_{\text{align eigen-axes}} x.$$

Because P is orthogonal, $P^\top$ is a rigid rotation/reflection that turns the orthonormal eigenvectors into the coordinate axes; D then stretches the i -th axis by the factor λ_i (reflecting it when $\lambda_i < 0$, collapsing it when $\lambda_i = 0$); and P rotates everything back. So a symmetric matrix is, geometrically, a **pure stretching along n mutually orthogonal directions** — its eigenvectors — with the eigenvalues as the stretch factors. It never shears. A generic (non-symmetric) matrix has no such orthogonal set of stretch axes; symmetry is exactly what buys the perpendicular ones.

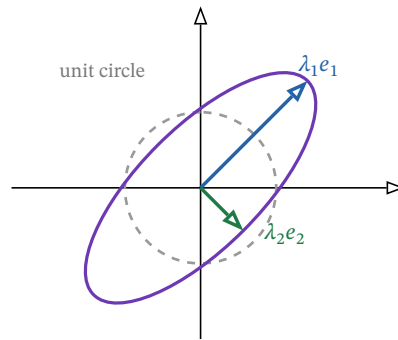


Figure 5: A symmetric matrix with eigenvalues $\lambda_1 > \lambda_2 > 0$ acting on the plane. The dashed **unit circle** is stretched into an **ellipse** (purple) whose axes lie along the orthonormal eigenvectors e_1, e_2 ; the semi-axis lengths are exactly the eigenvalues λ_1 and λ_2 . The matrix stretches space by λ_i along direction e_i and does nothing else — no shear. A negative eigenvalue would flip its axis; a zero one would flatten the ellipse onto a line.

6.5. The recipe, with a worked case

To orthogonally diagonalize a symmetric A :

1. Find the eigenvalues via $\chi_{A(\lambda)} = \det(A - \lambda I) = 0$ (all real).
2. For each eigenvalue, find a basis of its eigenspace. Eigenvectors from **distinct** eigenvalues are already orthogonal; within a **repeated** eigenvalue's eigenspace, run **Gram–Schmidt** to orthonormalize.
3. Normalize all eigenvectors and place them as the columns of P ; place the matching eigenvalues on the diagonal of D . Then $A = PDP^T$.

Example — a full orthogonal diagonalization. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (symmetric). Its characteristic polynomial is

$$\det(A - \lambda I) = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1),$$

so the eigenvalues are 3 and 1. Solving $(A - 3I)v = 0$ gives eigenvector $(1, 1)$, and $(A - I)v = 0$ gives $(1, -1)$. These are **orthogonal**, as the theory promised for distinct eigenvalues. Normalizing each to unit length:

$$P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Here P is orthogonal ($P^T P = I$), and one checks $A = PDP^T$. Because P is orthogonal, the inverse it would need in ordinary diagonalization is just its transpose — no matrix inversion is required.

Example — a repeated eigenvalue needing Gram–Schmidt. Let $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$, symmetric. Its characteristic polynomial factors as $\chi_A(\lambda) = -(\lambda - 4)(\lambda - 1)^2$, so the eigenvalues are $\lambda = 4$ (simple) and $\lambda = 1$ (repeated, multiplicity 2).

The simple eigenvalue. Solving $(A - 4I)v = 0$ gives the eigenvector $(1, 1, 1)$, which we normalize to $e_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$.

The repeated eigenvalue. Solving $(A - I)v = 0$ amounts to the single equation $v_1 + v_2 + v_3 = 0$, whose solution space E_1 is a **plane**. A natural basis read straight off that equation is

$$v_1 = (1, -1, 0), \quad v_2 = (1, 0, -1),$$

but these are **not** orthogonal: $\langle v_1, v_2 \rangle = 1 \neq 0$. This is the case the recipe warned about — we must run **Gram–Schmidt inside** E_1 .

Take $w_1 = v_1 = (1, -1, 0)$, with $\langle w_1, w_1 \rangle = 2$. Then

$$w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 = (1, 0, -1) - \frac{1}{2}(1, -1, 0) = \left(\frac{1}{2}, \frac{1}{2}, -1\right).$$

Now $\langle w_1, w_2 \rangle = \frac{1}{2} - \frac{1}{2} + 0 = 0$ ✓. Normalizing,

$$e_2 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad e_3 = \frac{w_2}{\|w_2\|} = \frac{1}{\sqrt{6}}(1, 1, -2),$$

using $\|w_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$.

Assemble. The three eigenvectors e_1, e_2, e_3 are orthonormal — e_1 is automatically orthogonal to the other two because it comes from a **distinct** eigenvalue, and $e_2 \perp e_3$ by the Gram–Schmidt step. So

$$P = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}, \quad D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

with P orthogonal and $A = PDP^T$. Without the Gram–Schmidt step the columns for $\lambda = 1$ would not have been orthogonal, and P would have failed $P^T P = I$ — the one place in the whole recipe where orthogonalization is not automatic.

7. Quadratic forms and positive definiteness

The spectral theorem has an immediate, high-value application: classifying **quadratic forms**. A symmetric matrix defines a function $q(x) = x^T Ax$, and the **signs of its eigenvalues** decide whether that function is a bowl, a dome, or a saddle. This is exactly the test used in optimization to classify critical points via the **Hessian**.

7.1. Quadratic forms

The map $q(x, y) = x^2 + y^2$ is the squared length in the plane; a mixed expression such as $q(x, y) = 2x^2 + 6xy + 5y^2$ is of the same kind. Each is homogeneous of degree two, and each is carried by a symmetric matrix — the object the definition records.

Definition 10. The **quadratic form** associated with a symmetric matrix $A \in \mathcal{M}_n(\mathbb{R})$ is the function

$$\begin{aligned} q : \mathbb{R}^n &\longrightarrow \mathbb{R} \\ x &\longmapsto x^T Ax = \sum_{i,j} a_{ij} x_i x_j \end{aligned}$$

Example — reading a form off its matrix. $q(x, y) = 2x^2 + 6xy + 5y^2$ corresponds to $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$: the diagonal entries are the coefficients of x^2, y^2 , and each **off-diagonal** entry is **half** the coefficient of the cross term xy (split symmetrically as $3 + 3 = 6$). Always halve the cross terms when building A — this is the single most common error.

7.2. Classification by eigenvalue signs

Diagonalizing $A = PDP^T$ and substituting $y = P^T x$ turns the form into a pure sum of squares $q = \sum \lambda_i y_i^2$ with **no cross terms** — the **principal axes**. The sign of each λ_i then determines everything.

Definition 11. A symmetric matrix A (and its quadratic form) is

- **positive definite** if $x^T Ax > 0$ for all $x \neq 0$;
- **positive semi-definite** if $x^T Ax \geq 0$ for all x ;
- **negative definite** / **negative semi-definite**: the reverse inequalities;
- **indefinite** if $x^T Ax$ takes both positive and negative values.

Theorem 5 (Eigenvalue classification). Let A be symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$. Then A is

- **positive definite** $\iff$ all $\lambda_i > 0$;
- **positive semi-definite** $\iff$ all $\lambda_i \geq 0$;
- **negative definite** $\iff$ all $\lambda_i < 0$;
- **negative semi-definite** $\iff$ all $\lambda_i \leq 0$;

- **indefinite** $\iff$ there are eigenvalues of **both** signs.

Proof. Write $q(x) = x^\top Ax = (P^\top x)^\top D(P^\top x) = \sum_i \lambda_i y_i^2$ where $y = P^\top x$. As x ranges over all non-zero vectors, so does y (since P is invertible). The sign behaviour of $\sum \lambda_i y_i^2$ is then dictated entirely by the signs of the λ_i : all $\lambda_i > 0$ makes it strictly positive, and all $\lambda_i < 0$ strictly negative; eigenvalues of both signs let it take both signs (choose y along the relevant axis), the indefinite case; and a zero eigenvalue lets $q = 0$ hold on a non-zero vector, which is what separates the semi-definite cases (all $\lambda_i \geq 0$, or all $\lambda_i \leq 0$) from the definite ones. $\square$

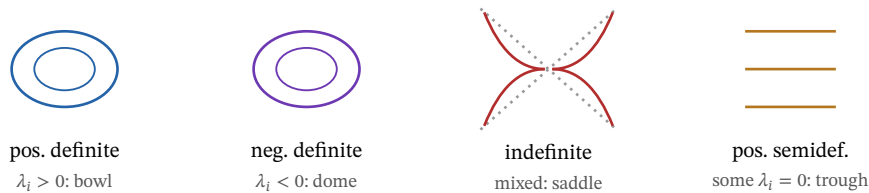


Figure 6: Level sets of a 2-variable quadratic form. Positive-definite (all $\lambda_i > 0$): nested ellipses around a **minimum** — a bowl. Negative-definite (all $\lambda_i < 0$): the same elliptical level sets, but around a **maximum** — a dome (the sign of the height, not the shape of the level curves, tells the two apart). Indefinite (mixed signs): hyperbolas — a saddle, going up one way and down another. Positive semi-definite (a zero eigenvalue): a flat trough with a direction of no curvature; negative semi-definite is its mirror image, an inverted trough.

7.3. The connection to the Hessian

Remark. In multivariable calculus, the second-order behaviour of a function f at a critical point is captured by its **Hessian matrix** H of second partial derivatives, which is **symmetric**. The classification above is exactly the **second-derivative test**: a positive-definite H gives a local **minimum** (the bowl), a negative-definite H a local **maximum** (the dome), and an indefinite H a **saddle** point (neither). The eigenvalue signs of the Hessian are how optimization algorithms tell minima from saddles.

Pitfall. Reading definiteness off the diagonal entries alone is wrong. The matrix $\begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ has positive diagonal yet eigenvalues 4 and -2 — it is **indefinite**. Positive diagonal entries are necessary but nowhere near sufficient; you must check the **eigenvalues** (or use a determinant-based criterion).

8. Orthogonal complement and projection

We turn to the second great theme: **approximation**. Given a subspace F and a vector v that may not lie in it, what is the point of F **closest** to v ? The answer rests on splitting any vector into a part inside F and a part exactly perpendicular to F — the **orthogonal complement** and the **orthogonal projection**.

8.1. The orthogonal complement

Fix a line through the origin in $\mathbb{R}^3$: the vectors at right angles to it fill the perpendicular plane. Fix a plane instead, and the vectors perpendicular to it fill a line. Collecting everything orthogonal to a given subspace always produces another subspace — the one the definition names.

Definition 12. Let F be a subspace of a Euclidean space E . Its **orthogonal complement** is

$$F^\perp = \{v \in E : \langle v, f \rangle = 0 \text{ for all } f \in F\},$$

the set of vectors orthogonal to **everything** in F . It is itself a subspace.

In practice F is given by a spanning set, and a vector lies in $F^\perp$ exactly when it is orthogonal to each spanning vector — a system of linear equations.

Example — computing an orthogonal complement. In $\mathbb{R}^3$ let $F = \text{Span}((1, 1, 0))$. A vector $(x, y, z) \in F^\perp$ iff $\langle (x, y, z), (1, 1, 0) \rangle = x + y = 0$. So $F^\perp = \{(x, -x, z)\} = \text{Span}((1, -1, 0), (0, 0, 1))$, a plane — the plane of all vectors perpendicular to the line F .

8.2. The direct sum decomposition

The central structural fact is that F and $F^\perp$ together fill all of E , with no overlap.

Theorem 6 (Orthogonal decomposition). For any subspace F of a Euclidean space E ,

$$E = F \oplus F^\perp, \quad \text{hence} \quad \dim F + \dim F^\perp = \dim E.$$

Every $v \in E$ has a **unique** decomposition $v = v_F + v_\perp$ with $v_F \in F$ and $v_\perp \in F^\perp$.

Proof. **No overlap:** if $w \in F \cap F^\perp$ then w is orthogonal to itself, $\langle w, w \rangle = 0$, so $w = 0$ by positive definiteness. **Spanning:** take an orthonormal basis $(e_1, \dots, e_k)$ of F (Gram–Schmidt guarantees one), and for any v set $v_F = \sum_i \langle v, e_i \rangle e_i \in F$. Then $v - v_F$ is orthogonal to every e_i , hence to all of F , so it lies in $F^\perp$. Thus $v = v_F + (v - v_F)$ is the required decomposition; the dimension formula follows from the direct sum. $\square$

Pitfall. The dimension formula $\dim F + \dim F^\perp = \dim E$ is the tool to **find** $\dim F^\perp$ without computing it directly: a plane in $\mathbb{R}^3$ ($\dim F = 2$) has a **line** as complement ($\dim F^\perp = 1$), and vice

versa. But it depends on the ambient E : the complement of a line is a 4-dimensional space inside $\mathbb{R}^5$, not the same as in $\mathbb{R}^3$. The complement is always taken **inside a stated ambient space**.

8.3. Orthogonal projection

The “ F -part” v_F of the decomposition is the orthogonal projection.

Definition 13. The **orthogonal projection** of v onto F is the unique $p_F(v) = v_F \in F$ from the decomposition above. If $(e_1, \dots, e_k)$ is an **orthonormal** basis of F , then

$$p_F(v) = \sum_{i=1}^k \langle v, e_i \rangle e_i.$$

Pitfall. The formula $p_F(v) = \sum \langle v, e_i \rangle e_i$ requires the basis to be **orthonormal**. With a merely orthogonal basis, divide each term by $\langle e_i, e_i \rangle$; with a general (non-orthogonal) basis the formula is simply wrong — orthonormalize first via Gram–Schmidt, or use the projection matrix below.

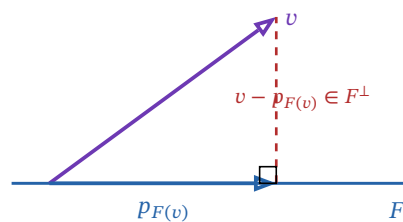


Figure 7: Orthogonal projection onto a subspace F . The vector v splits into $p_F(v) \in F$ (its shadow on F) and the perpendicular remainder $v - p_F(v) \in F^\perp$ (the right angle is marked). This is the unique way to write v as “something in F plus something perpendicular to F ”.

8.4. The projection matrix

When F is a subspace of $\mathbb{R}^n$, the projection is a linear map, hence a matrix.

Proposition 11 (Projection matrix). If the columns of Q form an **orthonormal** basis of $F \subset \mathbb{R}^n$, the orthogonal projection onto F is left-multiplication by

$$P_F = QQ^\top.$$

More generally, if the columns of M are **any** basis of F (not necessarily orthonormal), then

$$P_F = M(M^\top M)^{-1}M^\top.$$

In both cases P_F is symmetric and **idempotent** ($P_F^2 = P_F$): projecting twice is the same as projecting once.

Proof. Orthonormal case. Writing the columns of Q as the orthonormal basis $(e_1, \dots, e_k)$, the orthonormal-basis formula for p_F gives, for the column vector v ,

$$p_F(v) = \sum_{i=1}^k \langle v, e_i \rangle e_i = \sum_{i=1}^k e_i (e_i^\top v) = \left(\sum_{i=1}^k e_i e_i^\top \right) v = QQ^\top v,$$

since $\sum_i e_i e_i^\top$ is exactly the product $QQ^\top$. So the projection is left-multiplication by $P_F = QQ^\top$.

General case. Now let the columns of M be any basis of F . The projection $p_F(v)$ lies in F , so $p_F(v) = Mx$ for some coefficient vector x , and the defining property is that the residual $v - Mx$ lies in $F^\perp$ — that is, it is orthogonal to every column of M , which says $M^\top(v - Mx) = 0$. Hence

$$M^\top Mx = M^\top v.$$

The matrix $M^\top M$ is **invertible**: its columns being a basis makes them linearly independent, and if $M^\top Mx = 0$ then $\|Mx\|^2 = x^\top M^\top Mx = 0$, so $Mx = 0$ and therefore $x = 0$. Thus $x = (M^\top M)^{-1} M^\top v$ and

$$p_F(v) = Mx = M(M^\top M)^{-1} M^\top v,$$

giving $P_F = M(M^\top M)^{-1} M^\top$. (When the basis is orthonormal, $M = Q$ with $M^\top M = I$, and this collapses back to $QQ^\top$.) Finally P_F is symmetric — $(M(M^\top M)^{-1} M^\top)^\top = M(M^\top M)^{-1} M^\top$, using that $M^\top M$ is symmetric — and idempotent, since applying it to a vector already in F returns that vector unchanged, so $P_F^2 = P_F$. $\square$

Example — projecting onto a line. Project $v = (3, 4)$ onto the line $F = \text{Span}((1, 0))$ in $\mathbb{R}^2$. The unit vector is $e = (1, 0)$, so $p_{F(v)} = \langle v, e \rangle e = 3(1, 0) = (3, 0)$ — drop the second coordinate. The projection matrix is $P_F = ee^\top = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, and indeed $P_F(3, 4)^\top = (3, 0)^\top$. The remainder $(0, 4) \in F^\perp$.

9. Best approximation, distance, and least squares

Everything assembled now yields the theorem the course was built for: the orthogonal projection of v onto F is the **closest point of F to v** . This is the **best-approximation theorem**, and its distance is the residual that least-squares regression minimizes.

9.1. The best-approximation theorem

Drop a perpendicular from a point onto a line in the plane: the foot of that perpendicular is the point of the line closest to the given point, and every other point of the line is strictly farther away. The theorem below is exactly this picture, raised to any subspace of a Euclidean space.

Theorem 7 (Best approximation). Let F be a subspace of a Euclidean space and v any vector. The orthogonal projection $p_{F(v)}$ is the **unique closest point of F to v** : for every $f \in F$ with $f \neq p_{F(v)}$,

$$\|v - p_{F(v)}\| < \|v - f\|.$$

Proof. For any $f \in F$, write $v - f = (v - p_{F(v)}) + (p_{F(v)} - f)$. The first bracket lies in $F^\perp$ and the second in F , so they are orthogonal, and Pythagoras gives

$$\|v - f\|^2 = \|v - p_{F(v)}\|^2 + \|p_{F(v)} - f\|^2.$$

The second term is ≥ 0 , and is zero only when $f = p_{F(v)}$. Hence $\|v - f\|$ is strictly larger than $\|v - p_{F(v)}\|$ unless $f = p_{F(v)}$. $\square$

The proof is the whole idea in one line: **the error $v - p_{F(v)}$ is perpendicular to F , and a perpendicular dropped to a subspace is shorter than any slanted line to it.**

9.2. Distance to a subspace

Definition 14. The **distance** from v to a subspace F is the minimal distance to its points, achieved at the projection:

$$d(v, F) = \|v - p_{F(v)}\| = \|v_\perp\|.$$

Example — distance from a point to a plane. Find the distance from $v = (1, 1, 1)$ to the plane $F = \text{Span}((1, 0, 0), (0, 1, 0))$ (the xy -plane) in $\mathbb{R}^3$. The projection drops the third coordinate: $p_{F(v)} = (1, 1, 0)$. The remainder is $v - p_{F(v)} = (0, 0, 1) \in F^\perp$, so

$$d(v, F) = \|(0, 0, 1)\| = 1.$$

The closest point of the plane is $(1, 1, 0)$, directly “below” v .

9.3. Why this is least-squares regression

Remark. Least-squares regression fits a model to data by minimizing the sum of squared errors — and that is **exactly** a distance minimization in a Euclidean space. To solve an overdetermined system $Ax \approx b$ (more equations than unknowns, no exact solution), we seek the x making Ax as close to b as possible. The achievable vectors Ax form the column space $F = \text{Im}(A)$, so the best Ax is the **orthogonal projection of b onto F** . The error $b - Ax$ must be perpendicular to F , i.e. $A^T(b - Ax) = 0$, which rearranges to the **normal equations**

$$A^T A x = A^T b.$$

Least-squares regression **is** the best-approximation theorem, and the regression line is the orthogonal projection of the data onto the space of fitted values.

Example — a least-squares line, computed. Fit a line $y = a + bx$ to the three data points $(0, 1)$, $(1, 2)$, $(2, 2)$. No line passes through all three, so we solve in the least-squares sense. Writing one row per data point, the model $y = a + bx$ becomes $A \begin{pmatrix} a \\ b \end{pmatrix} \approx b$ with

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

The normal equations $A^T A \begin{pmatrix} a \\ b \end{pmatrix} = A^T b$ use

$$A^T A = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad A^T b = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Solving $3a + 3b = 5$ and $3a + 5b = 6$ gives $b = 1/2$ and $a = 7/6$, so the best-fit line is

$$y = \frac{7}{6} + \frac{1}{2}x.$$

Its residuals at the three points are $-1/6, +2/6, -1/6$, which sum to 0 — the sign that the error vector is orthogonal to the fitted values, exactly as the best-approximation theorem predicts.

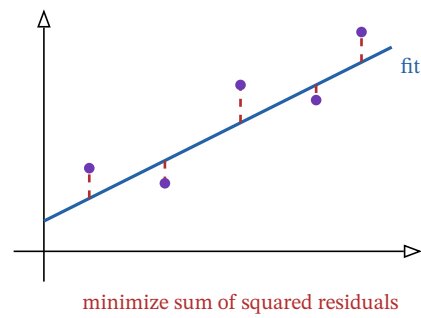


Figure 8: Least-squares regression as best approximation. The dashed residuals are the gaps between data (purple) and the fitted line (blue). Minimizing the sum of their squares is exactly projecting the data vector b orthogonally onto the column space of the design matrix — the closest reachable point, with error perpendicular to that space.